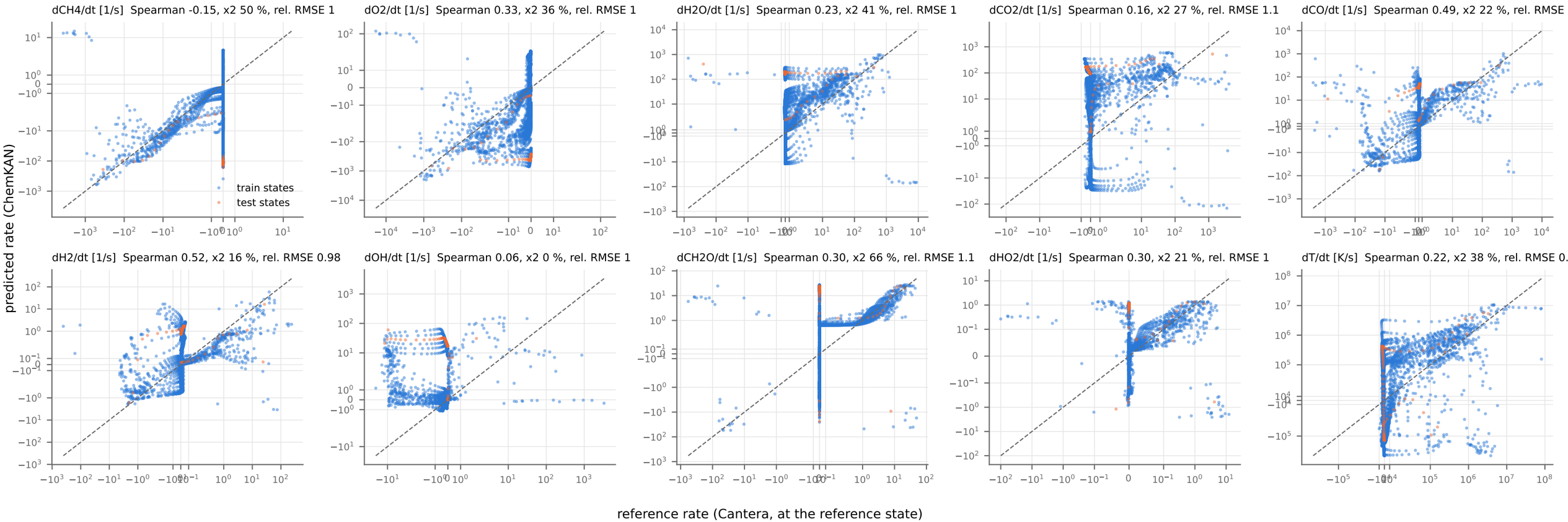

methane: instantaneous rates at every state of the reference trajectories (symlog axes). Training states: rank correlation, share within a factor 2 (states with |rate| > 1 % of the species' max), relative RMSE
dCH4/dt [1/s] Spearman -0.15, x2 50 %, rel. RMSE 1
dO2/dt [1/s] Spearman 0.33, x2 36 %, rel. RMSE 1
dH2O/dt [1/s] Spearman 0.23, x2 41 %, rel. RMSE 1
dCO2/dt [1/s] Spearman 0.16, x2 27 %, rel. RMSE 1.1
dCO/dt [1/s] Spearman 0.49, x2 22 %, rel. RMSE
dH2/dt [1/s] Spearman 0.52, x2 16 %, rel. RMSE 0.98
dOH/dt [1/s] Spearman 0.06, x2 0 %, rel. RMSE 1
dCH2O/dt [1/s] Spearman 0.30, x2 66 %, rel. RMSE 1.1
dHO2/dt [1/s] Spearman 0.30, x2 21 %, rel. RMSE 1
dT/dt [K/s] Spearman 0.22, x2 38 %, rel. RMSE 0.
train states
test states
predicted rate (ChemKAN)
reference rate (Cantera, at the reference state)